EMMA FANUELE

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EDUCATION

Columbia University Mailman School of Public Health, New York, NY — Expected May 2027
Master of Public Health, Epidemiology; Certificate in Applied Biostatistics and Public Health Data Science; **GPA: 4.10**
Relevant Coursework: Applied Regression I–II, Analysis of Categorical Data, Introduction to Randomized Controlled Trials, Epidemiology II–III
Master's Thesis (in progress): Public perceptions of antimicrobial resistance — designing and leading the analysis of survey data from the AMR/UTI study (see Research Experience)

McGill University, Montreal, QC — Jun 2025
Bachelor of Arts, Honours Faculty Program in Environment, Ecological Determinants of Health in Society; Dual Minor: Statistics, Field Studies; **GPA: 3.92**
Relevant Coursework: Sampling Theory & Applications, advanced Probability & Statistics, intermediate Calculus & Linear Algebra; Panama Field Study Semester (2024); Africa Field Study Semester (2023, Kenya & Tanzania)
Honours Thesis: Assessing the influence of the mangrove ecosystem on food security (Advisor: Prof. Hugo Melgar-Quiñonez)

SKILLS

Programming & Tools – R (advanced); SAS, incl. SUDAAN (intermediate); Python (foundational); SQL (foundational; PROC SQL in SAS); LaTeX; Microsoft Suite
Certifications – CITI Program Biomedical Research Course; Tri-Council Policy Statement: Ethical Conduct for Research Involving Humans (TCPS 2); HIPAA Privacy and Security Essentials Course; Mindgrove MedTech Essentials Course
Languages – English (native); Spanish (working proficiency)

RESEARCH EXPERIENCE

Columbia University Irving Medical Center, New York, NY
Cross-National Analysis of Suicide Methods Across Decades — *Jul 2026 – Present*
Research Assistant (Dr. Tomer Levy and Dr. Larkin McReynolds)
- Lead the statistical design of a new cross-national study comparing suicide methods across decades and countries

Cross-National Analysis of Firearm Access and Suicide Rates — *Oct 2025 – Oct 2026*
Research Assistant (Dr. Tomer Levy and Dr. Larkin McReynolds)
- Conducted all statistical analyses investigating associations between civilian firearm prevalence and suicide rates across 62 countries (2017–2021), using negative binomial regression with cluster-robust standard errors in R
- Harmonized multi-year, sex-stratified WHO and UN demographic datasets (20,000+ observations) into a reproducible pipeline for data cleaning, merging, and linkage to firearm ownership indicators
- Conducted interaction analyses examining firearm availability by age and sex to identify subgroup differences in firearm suicide risk
- Produced publication-ready figures in ggplot2 (dot plots, scatter plots, age-stratified line plots)
- Finalizing the manuscript for publication; first author on an accepted oral presentation at a national conference (see Presentations)

Treatment Patterns in Patients With Eosinophilic Esophagitis Referred for Expert Consultation — *Jan 2026 – Aug 2026*
Research Assistant (Dr. Sydney Pomenti and Dr. David Katzka)
- Conducted statistical analyses in R (gtsummary) on cross-sectional survey data from 52 eosinophilic esophagitis experts, assessing provider characteristics, referral patterns, perceived causes of treatment failure, diagnostic approaches, and prior therapies
- Finalized the manuscript's reported statistics, tables, and figures; co-authored the resulting paper, published in Alimentary Pharmacology & Therapeutics (2026)

Columbia University Mailman School of Public Health, New York, NY
SOLER Study of Student Engagement in the Core Curriculum — *Sep 2026 – Present*
Student Research Worker (Heilbrunn Department of Population and Family Health & Digital Learning Studio)
- Merge, clean, and analyze de-identified quantitative data in SPSS according to the study's data analytic plan, producing tabular and narrative summaries and data visualizations
- Set up participant surveys, coordinate study groups, and post engagement nudges in Canvas; assist with data downloads
- Participate in regular study team meetings and contribute to dissemination of findings, including manuscript writing and revising

Bassett Healthcare Network, Cooperstown, NY
NYCAMH Research Practicum *Jun 2026 – Jul 2026*
Research Intern (Dr. Amanda Roome and Dr. Thelma Mielenz)

- Analyzed New York State Department of Health county-level tick-borne disease incidence and tick pathogen positivity data, producing R visualizations across eight counties and addressing small-denominator reporting challenges
- Contributed to a randomized controlled trial evaluating tick-control interventions on agricultural properties through literature reviews and manuscript drafting and editing
- Researched Alpha-Gal Syndrome and the elevated risk faced by farmers; created an informational brochure on available resources for farmers affiliated with NYCAMH
- Supported occupational health research on structural barriers affecting agricultural worker health and safety through literature reviews
- Developed a poster and abstract uniting tick-borne disease surveillance, the tick-control RCT, and Alpha-Gal Syndrome education in a single public health narrative (see Presentations)

McGill University, Montreal, QC
Antimicrobial Resistance in Relation to Urinary Tract Infections *May 2023 – May 2025*
Research Assistant (Dr. Lysanne Campeau, McGill University; Dr. Lori Lerner, VA Boston Healthcare System)

- Collaborated with urologist mentors to research antimicrobial resistance (AMR) in relation to UTIs
- Designed and implemented a survey to gauge public perceptions of the antibiotic resistance crisis and behaviors contributing to AMR (199 completed surveys)
- Conducted linear regression analyses, finding that participants were generally unaware of the AMR crisis and took few specific actions to avoid resistance
- Presented study findings as a poster at the 2025 American Urological Association Annual Meeting (see Presentations); contributing to a manuscript in preparation
- Now extending this dataset into my master's thesis, designing and leading the analysis (see Education)

Mangroves and Food Security Honours Thesis Project *Aug 2024 – Apr 2025*
Honours Student (Professor Hugo Melgar-Quiñonez)

- Conducted a literature review of 50+ publications analyzing the impact of mangrove ecosystems on human nutritional health and food security; presented findings as a poster (see Presentations)

TECHO Panamá, Panama City, Panama
Water & Citizenship: Experiences of Water Access in Nueva Luz N°2 *Jan 2024 – Apr 2024*
Research Intern

- Assessed water insecurity in an informal settlement through 20 stakeholder interviews
- Identified gaps in water access, storage capacity, and citizenship rights awareness, highlighting the health, economic, and social impacts
- Developed a workshop and pamphlet to support dialogue on citizenship rights and the sustainable implementation of rainwater capture systems

VA Boston Healthcare System, Boston, MA
VA Boston Research Student Summer Institute (VARSSI) *May 2024 – Aug 2024*
Research Intern

- Analyzed U.S. electronic health record (EHR) data in collaboration with Boston Scientific to assess incontinence rates following benign prostatic hyperplasia surgery
- Managed project planning, including endpoint design and drafting statistical exhibits
- Found that post-surgical incontinence rates decreased over time, and characterized the limitations of real-world data for determining symptom resolution

Tufts University, Boston, MA
Barriers to Mind-Body Movement (MBMI) Systematic Review *Jun 2024 – Feb 2025*
Research Assistant

- Screened 2,000 abstracts in Covidence to select full articles on the influence of mind-body movement interventions (yoga, Tai Chi, Qi Gong) on health outcomes
- Reviewed 60 full-text articles and assisted with data extraction; co-authored the published protocol (see Publications)

PUBLICATIONS

- Pomenti, S. F., Stoffel, E., Choy, A., **Fanuele, E.**, Gonsalves, N., Dellon, E. S., & Katzka, D. A. (2026). Treatment patterns in patients with eosinophilic oesophagitis referred for expert consultation. *Alimentary Pharmacology & Therapeutics*. Advance online publication. https://doi.org/10.1111/apt.70951
- Wexler, R. S., Joyce, C. T., Reichman, R., Pereira, C., **Fanuele, E.**, Hurstak, E., Laird, L. D., Lavretsky, H., Wang, C., Saper, R., Alcorn, K. S., Mittman, B. S., & Roseen, E. J. (2026). Determinants of Qi Gong, Tai Chi, and Yoga use for health conditions: A systematic review protocol. *Global Advances in Integrative Medicine and Health, 15*. https://doi.org/10.1177/27536130261417682

PRESENTATIONS

** Presenting author*

- **Fanuele, E.***, McReynolds, L., Hoven, C., Lin, S., & Levy, T. (2026, December). *National firearm holdings and firearm suicide: A cross-national analysis by age and sex* [Oral presentation]. National Research Conference for the Prevention of Firearm-Related Harms, Baltimore, MD. (Accepted)
- **Fanuele, E.***, & Roome, A. (2026, November). *Rising and emerging tick-borne disease burden in northeastern US rural agricultural counties* [Poster presentation]. Epidemiology Master's Student Day, Columbia University Mailman School of Public Health, New York, NY. (Upcoming)
- Helrich, S., **Fanuele, E.***, Campeau, L., & Lerner, L. (2025, April). *Attitudes towards antibiotic prescribing and awareness of antimicrobial resistance in Americans and Canadians* [Poster presentation]. American Urological Association Annual Meeting, Las Vegas, NV.
- **Fanuele, E.***, & Melgar-Quiñonez, H. R. (2025, March). *Assessing the influence of the mangrove ecosystem on food security* [Poster presentation]. Bieler School of Environment Honours Symposium, McGill University, Montreal, QC.

WORK EXPERIENCE

Columbia University Mailman School of Public Health, New York, NY
Quantitative Methods of Public Health *Sep 2026 – Dec 2026*
Teaching Assistant

- Lead weekly labs for a section of 17 first-year MPH students – many of them physicians and other mid-career professionals – progressing from foundational statistics through regression analysis and interpretation
- Hold weekly office hours, grade weekly homework assignments and labs, and attend lectures to provide technical support

New York City Department of Health – Bureau of Epidemiology Services, New York, NY
HRTP: A Public Health Internship (Survey Data and Analysis Unit) *Jun 2026 – Aug 2026*
Data Analyst Intern (Wen Qin Deng, MPH)

- Conducted small-area population estimation for New York City's Neighborhood Tabulation Areas in SAS, testing and comparing raking methodologies for neighborhood-level demographic surveillance and public-use dataset preparation
- Processed and validated complex survey data using SAS/SUDAAN, including convergence checking and missing data assessment
- Presented the methodology comparison and results to the unit at the end of the internship, recommending next steps

VOLUNTEER WORK EXPERIENCE

Pothawira International, Houston, TX; Salima, Malawi *Jun 2018 – Present*
Fundraising and Operations Volunteer; Medical Volunteer

- Fundraise for the Pothawira clinic, school, and orphanage; climbed Mount Kilimanjaro in 2023 to raise funds for the clinic's surgical center
- Organized a Malawian art auction showcasing Malawian culture through food and dance (Feb 2019)
- Scribed for physicians in the clinic; obtained blood samples for malaria and hemoglobin testing

World Food Programme, Seoul, South Korea *Jun 2022 – Aug 2022*
Communications, Advocacy, and Marketing Intern

- Advocated for sustainable food systems to reduce the environmental impact of, and waste from, Korean food systems
- Completed a final presentation and write-up on promoting sustainable food systems through advocacy for animal welfare and eating disorder resources in Korea

ACTIVITIES

Community Public Health Literacy (CIPHER), Columbia University *Aug 2025 – Present*
Programming and Events Lead

- Plan, coordinate, and execute events and programming, including workshops and community-centered programs
- Volunteer at local community events distributing supplies to 20+ families with children in Washington Heights; support sports-focused programming for local women and girls
- Tutor local elementary-school kids in math and reading in a before-school program

Mailman Jewish Student Association, Columbia University *Aug 2025 – Present*
Co-President

- Plan and coordinate events celebrating Jewish culture, creating a space for Jewish students and those interested in the culture to connect
- Coordinate events with other Jewish groups at Columbia, including the Grad Network and the CUIMC Jewish Cultural Employee Resource Group

McGill Cheerleading, McGill University *Sep 2022 – Mar 2025*
Assistant Coach; Athlete; Team Events Coordinator
- Coached the Level 4 team, organizing practices and teaching new skills to 14 athletes
- Competed as a Level 7 athlete with 12 years of experience in tumbling and flying; organized team bonding events and assisted with coaching Level 7 while injured

International Student Buddy Program, McGill University *Aug 2024 – Dec 2024*
Mentor
- Mentored two Singaporean exchange students, answering questions and providing resources

SELECTED AWARDS AND HONORS

First-Class Honours; Dean's Honour List (2021–2025, all four years); Golden Key International Honour Society; Field Studies Mobility Award — McGill University